Bank Marketing Campaign Analysis

Portfolio Case Study

Tableau | Data Viz | Banking / FinTech | Marketing Analytics

Problem Statement & Stakeholder Context

A Portuguese retail bank ran a series of direct telephone marketing campaigns to promote term deposit subscriptions. Despite reaching over 45,000 clients, the overall subscription rate was a modest **11.70%** — highlighting a critical need to better understand *who* subscribes, *when* to call, and *which* client segments warrant prioritisation.

The marketing team needed an interactive reporting layer to move beyond flat spreadsheets and support data-driven decisions around campaign timing, targeting criteria, and budget allocation.

Dataset Source & Scope

Source	UCI Machine Learning Repository — Bank Marketing Dataset
Records	45,211 client contacts across multiple campaign cycles
Features	17 variables including demographics, financials, contact history, and campaign metadata
Target Variable	Term deposit subscription (binary: yes / no)

Data Cleaning & Transformation

- Categorised client **age** into six bands (18–24, 25–34, 35–44, 45–54, 55–64, 65+) for cohort-level comparison.
- Defined four **balance bands** (negative_or_zero, low, medium, high) to segment clients by financial standing.
- Mapped raw job codes to human-readable labels and grouped rare categories where needed.

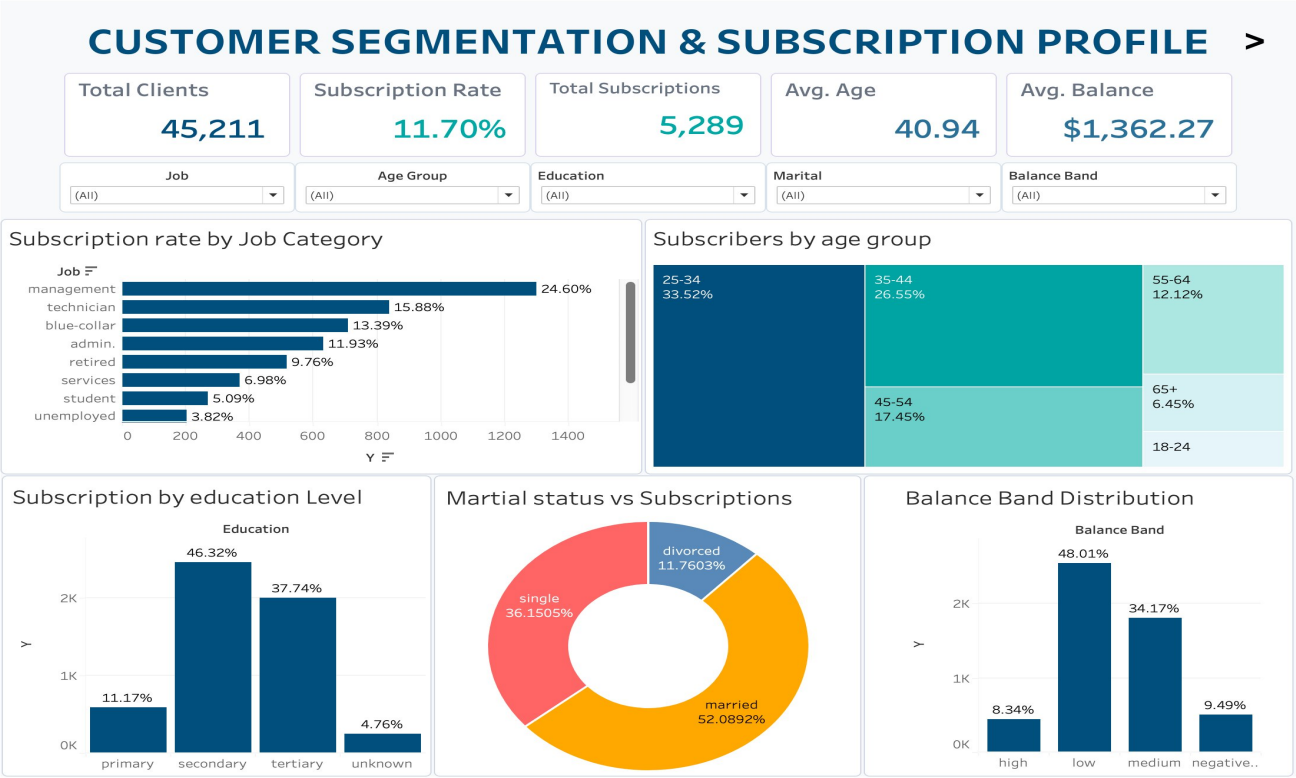
- Derived a **previously contacted** flag from the *pdays* field to isolate warm vs. cold outreach effects.
- Standardised missing values encoded as 'unknown' and validated referential integrity across campaign tables.
- Built calculated fields in Tableau for subscription rate, contact-to-conversion ratio, and lift metrics.

KPI Framework

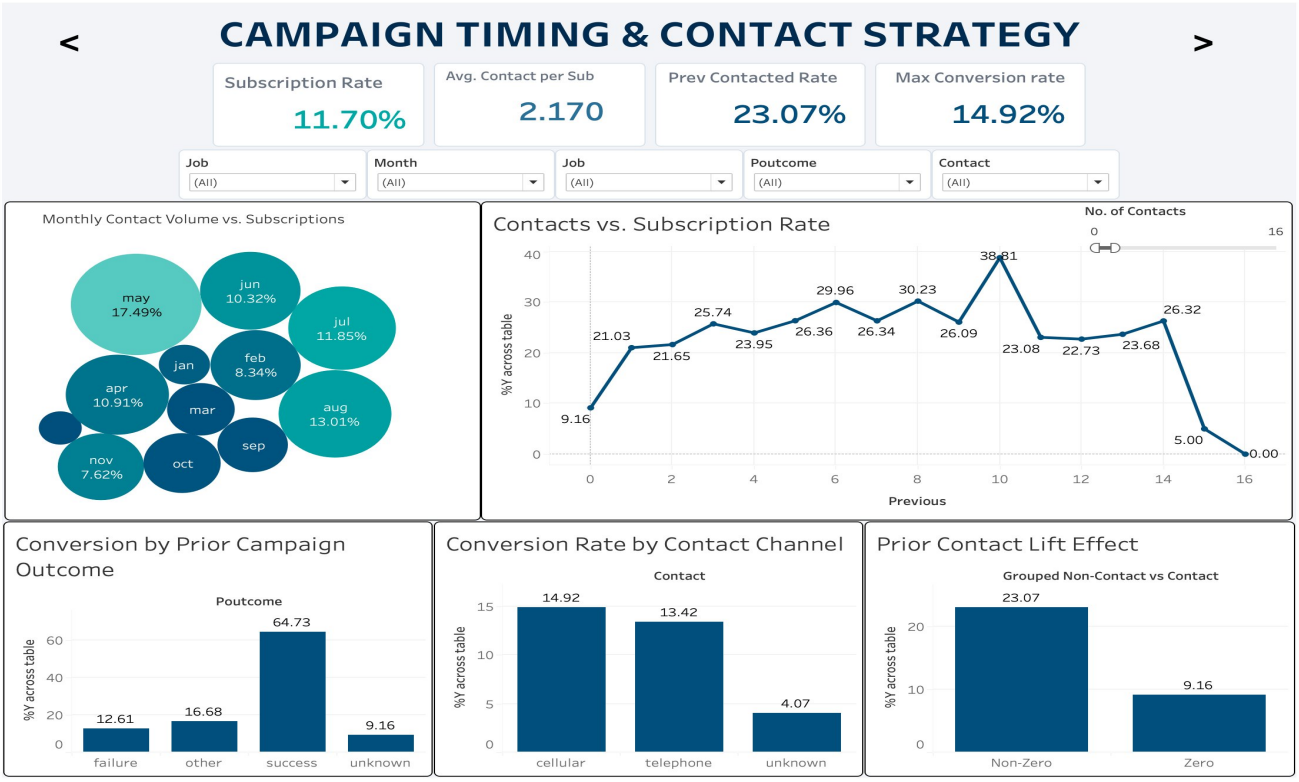
45,211 Total Clients	**11.70%** Subscription Rate	**5,289** Total Subscriptions	**41 yrs** Avg. Client Age
€1,362 Avg. Yearly Balance	**2.17** Avg. Contacts / Client	**23.07%** Previously Contacted Rate	**14.92** Max Contact Conversion Rate

Tableau Dashboard

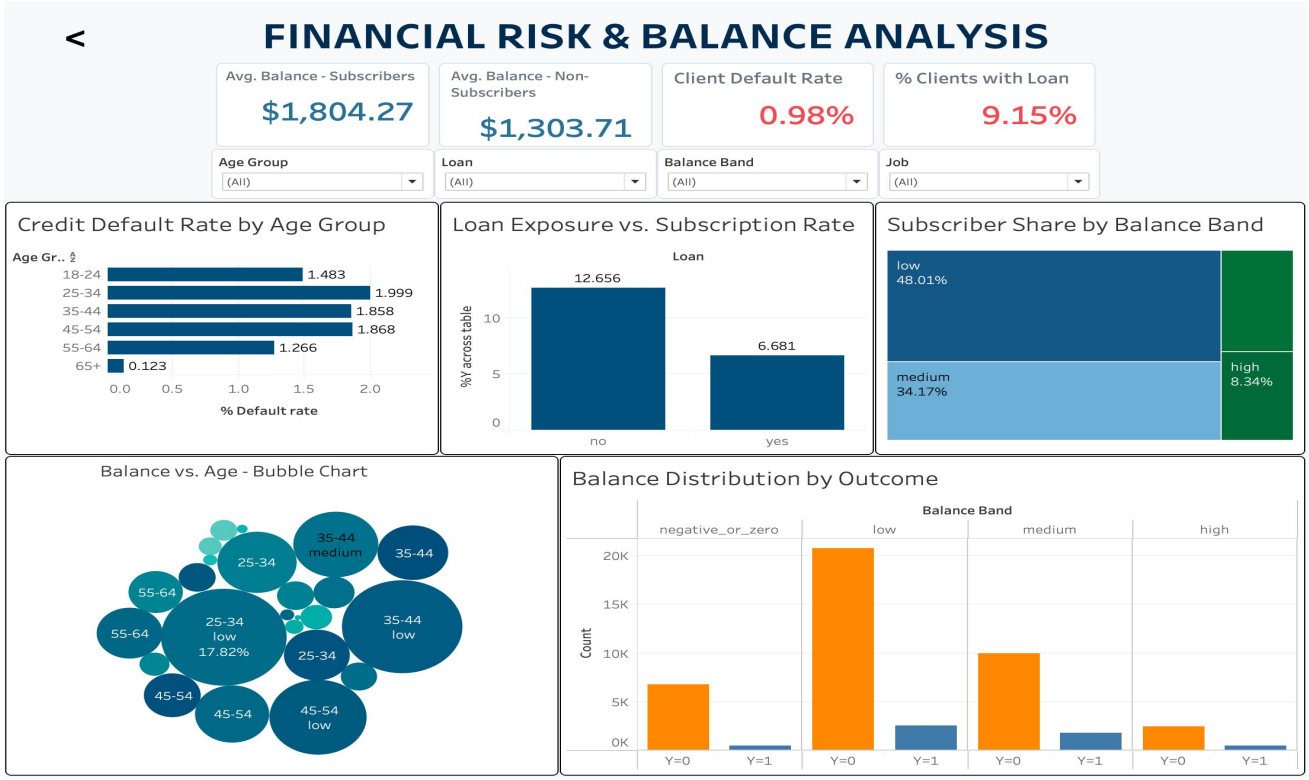
Three dashboard pages provide interactive exploration of client demographics, campaign contact behaviour, and financial risk profiling.



Page 1 — Client Demographics & Subscription Drivers



Page 2 — Campaign Contact Behaviour & Conversion



Page 3 — Financial Risk & Balance Profiling

Key Insights

1 **Management leads conversion**

Management-role clients show the highest subscription rate at 24.60%, nearly double the dataset average. Technicians (15.88%) and blue-collar workers (13.39%) follow, while self-employed and unemployed segments trail below 4%.

2 **The 25–34 cohort is the largest subscriber group**

33.52% of all subscribers fall in the 25–34 age band, followed closely by 35–44 (26.55%). The youngest (18–24) and oldest (65+) cohorts together account for only ~10%, suggesting the campaign resonates most with early-career professionals.

3 **May is the highest-volume contact month but not peak efficiency**

May sees the most contacts (17.49% of subscriptions), yet the Contacts vs. Subscription Rate chart reveals that ~10 prior contacts yields the peak conversion rate of 38.81% — heavy contact volumes alone do not guarantee the best outcome.

4 **Prior campaign success is a powerful signal**

Clients with a successful prior campaign outcome convert at 64.73%, versus 16.68% for failures and 12.61% for no prior outcome. Warm re-engagement with previously successful clients is the single strongest conversion lever available.

5 **Telephone outperforms cellular for conversion**

The telephone channel achieves a 14.92% conversion rate compared to 13.42% for cellular and only 4.07% for an alternative channel — pointing to clear channel prioritisation opportunities for future campaigns.

6 **Having a loan halves subscription likelihood**

Clients with an existing loan subscribe at 6.68%, while those without subscribe at 12.66% — a near 2× difference. Loan exposure is a meaningful negative predictor and should be factored into targeting filters.

Recommendations & Expected Impact

→ **Prioritise warm leads.** Build a re-engagement queue of clients with prior successful outcomes. This segment converts at 64.73% — deploying even a fraction of campaign budget here would materially lift overall subscription numbers.

→ **Segment by job and age.** Focus acquisition spend on management-role clients aged 25–44. This intersection covers the highest-probability segments and enables more personalised messaging.

→ **Optimise contact frequency.** The 10-contact sweet spot (38.81% conversion) suggests diminishing returns beyond this threshold. Capping contacts and reallocating capacity to new prospects would improve efficiency.

→ **Shift budget to telephone channel.** Telephone outperforms cellular by ~1.5 pp. Even modest reallocation from lower-performing channels adds up across 45,000+ contacts.

→ **Filter out high loan-exposure clients.** Clients with existing loans show substantially lower conversion. Applying a loan-flag exclusion or de-prioritisation rule would sharpen targeting precision.

→ **Time campaigns around peak months.** May, July, and August show the strongest subscription rates. Aligning budget peaks with these months maximises reach during high-intent windows.